

Question Bank — Programming & Data Structures

Module 4 companion | PYQ-style + Practice | exam-tagged, answers & explanations

■ How to use this bank

Two sections: PYQ-style (modelled on real IBPS/SSC/GATE/PSU/DSSSB questions, exam-tagged) and **Practice** (full coverage by sub-topic).

Every question has answer + explanation. **Tags:** IBPS • Coal India • DSSSB • All 3 • GATE-level • SSC/other.

For IBPS: prioritise IBPS / All-3; GATE-tagged (pointer/recursion tracing, tree/heap operations) are Coal-India depth.

SECTION A — PYQ-style (modelled on actual exam questions)

Programming fundamentals

Q1

All 3 exams

Which translator converts the entire program at once and reports all errors together?

- (a) Interpreter
- (b) Compiler
- (c) Assembler
- (d) Linker

Answer: (b)

A compiler translates the whole program in one pass; an interpreter goes line by line.

Q2

All 3 exams

SSC/other

Which language is executed line-by-line by an interpreter?

- (a) C
- (b) Python
- (c) Assembly
- (d) COBOL

Answer: (b)

Python is interpreted; C is compiled.

Q3

All 3 exams

In pass-by-value, changes to a parameter inside a function:

- (a) Affect the original
- (b) Do not affect the original
- (c) Delete the variable
- (d) Cause an error

Answer: (b)

Pass-by-value copies the argument; the original is unchanged. Pass-by-reference can modify the original.

Q4

All 3 exams

A while loop checks its condition:

- (a) After the body (runs ≥ 1)
- (b) Before the body (may run 0 times)
- (c) Never
- (d) Only once

Answer: (b)while tests first (0+ executions); do-while tests after (≥ 1 execution).**Q5**

IBPS SO IT

All 3 exams

Recursion requires which of these to terminate?

- (a) A loop
- (b) A base case
- (c) A pointer
- (d) Global variable

Answer: (b)

Every recursion needs a base case; without it, infinite recursion overflows the stack.

Q6

GATE-level

Coal India MT

Recursion uses which data structure internally to manage calls?

- (a) Queue
- (b) Stack
- (c) Heap
- (d) Tree

Answer: (b)

The call stack stores each recursive call's frame (LIFO).

OOP

Q7

All 3 exams

Which is NOT one of the four OOP pillars?

- (a) Encapsulation
- (b) Inheritance
- (c) Polymorphism
- (d) Compilation

Answer: (d)

The pillars are Encapsulation, Abstraction, Inheritance, Polymorphism.

Q8

IBPS SO IT

All 3 exams

Bundling data and methods while hiding internal state is:

- (a) Inheritance
- (b) Encapsulation
- (c) Polymorphism
- (d) Abstraction

Answer: (b)

Encapsulation bundles data + methods and restricts direct access (private members).

Q9

All 3 exams

SSC/other

Method overloading (same name, different parameters) is resolved at:

- (a) Run time
- (b) Compile time
- (c) Link time
- (d) Never

Answer: (b)

Overloading is compile-time (static) polymorphism; overriding is run-time (dynamic).

Q10

DSSSB TGT

All 3 exams

An object is:

- (a) A blueprint
- (b) An instance of a class
- (c) A function
- (d) A data type only

Answer: (b)

A class is the blueprint; an object is an instance of that class.

Q11

DSSSB TGT

All 3 exams

Inheritance represents which relationship?

- (a) has-a
- (b) is-a
- (c) uses-a
- (d) part-of

Answer: (b)

Inheritance models an 'is-a' relationship (a Dog is-a Animal).

Q12

Coal India MT

GATE-level

Java avoids multiple inheritance of classes (diamond problem) by using:

- (a) Structures
- (b) Interfaces
- (c) Pointers
- (d) Friend functions

Answer: (b)

Java uses interfaces instead of multiple class inheritance to avoid ambiguity.

Linear data structures

Q13

All 3 exams

Which offers $O(1)$ random access by index?

- (a) Linked list
- (b) Array
- (c) Stack
- (d) Queue

Answer: (b)

Arrays are contiguous, so `element[i]` is $O(1)$. Linked lists need $O(n)$ traversal.

Q14

All 3 exams

A stack follows which order?

- (a) FIFO
- (b) LIFO
- (c) Priority
- (d) Random

Answer: (b)

Stack = Last-In-First-Out (push/pop at the top).

Q15

All 3 exams

SSC/other

A queue follows which order?

- (a) LIFO
- (b) FIFO
- (c) Priority
- (d) Random

Answer: (b)

Queue = First-In-First-Out (enqueue at rear, dequeue at front).

Q16

IBPS SO IT

All 3 exams

Which data structure is used to convert infix to postfix?

- (a) Queue
- (b) Stack
- (c) Tree
- (d) Graph

Answer: (b)

Infix→postfix conversion and postfix evaluation use a stack.

Q17

All 3 exams

Function calls during program execution are managed by a:

- (a) Queue
- (b) Stack
- (c) Heap
- (d) Hash table

Answer: (b)

The call stack manages nested function calls (LIFO).

Q18

DSSSB TGT

All 3 exams

In a circular queue, the advantage over a linear queue is:

- (a) More memory
- (b) Reuse of freed front space
- (c) Faster CPU
- (d) Random access

Answer: (b)

A circular queue reuses vacated front slots, avoiding false 'full' conditions.

Q19

All 3 exams

Insertion/deletion at both ends is supported by a:

- (a) Stack
- (b) Deque
- (c) Priority queue
- (d) Simple queue

Answer: (b)

A deque (double-ended queue) allows insert/delete at both front and rear.

Trees, heaps & graphs

Q20

All 3 exams

Inorder traversal of a Binary Search Tree yields nodes in:

- (a) Random order
- (b) Sorted (ascending) order
- (c) Reverse order
- (d) Level order

Answer: (b)

Inorder (Left-Node-Right) of a BST produces ascending sorted output.

Q21

IBPS SO IT

All 3 exams

Preorder traversal visits nodes in which order?

- (a) Left-Node-Right
- (b) Node-Left-Right
- (c) Left-Right-Node
- (d) Right-Node-Left

Answer: (b)

Preorder = Node → Left → Right (NLR).

Q22

All 3 exams

SSC/other

Which traversal is used to delete a tree (children before parent)?

- (a) Inorder
- (b) Preorder
- (c) Postorder
- (d) Level-order

Answer: (c)

Postorder (Left-Right-Node) processes children before the parent — safe for deletion.

Q23

All 3 exams

In a max-heap, the largest element is at the:

- (a) Leaf
- (b) Root
- (c) Middle
- (d) Last node

Answer: (b)

A max-heap keeps the maximum at the root (parent $\geq$ children).

Q24

All 3 exams

Level-order traversal of a tree uses a:

- (a) Stack
- (b) Queue
- (c) Heap
- (d) Hash

Answer: (b)

Level-order (BFS) uses a queue.

Q25

GATE-level

Coal India MT

In a BST, average time for search is:

- (a) $O(1)$
- (b) $O(\log n)$
- (c) $O(n)$
- (d) $O(n \log n)$

Answer: (b)Balanced BST search is $O(\log n)$ average; worst case $O(n)$ for a skewed tree.**Q26**

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GATE-level

The maximum number of nodes at level L of a binary tree (root at level 0) is:

- (a) L
- (b) $2L$
- (c) 2^L
- (d) L^2

Answer: (c)Max nodes at level $L = 2^L$.**Q27**

All 3 exams

BFS graph traversal uses a queue; DFS uses a:

- (a) Queue
- (b) Stack (or recursion)
- (c) Heap
- (d) Priority queue

Answer: (b)

DFS uses a stack (or recursion); BFS uses a queue.

Hashing

Q28

All 3 exams

Hashing gives which average-case lookup time?

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(1)$
- (d) $O(n \log n)$

Answer: (c)A good hash table gives $O(1)$ average search/insert/delete.

Q29

IBPS SO IT

All 3 exams

When two keys map to the same index, it is called a:

- (a) Overflow
- (b) Collision
- (c) Fault
- (d) Miss

Answer: (b)

A collision; resolved by chaining or open addressing.

Q30

All 3 exams

SSC/other

Which is an open-addressing collision-resolution method?

- (a) Chaining
- (b) Linear probing
- (c) Linked list per slot
- (d) Bucket lists

Answer: (b)

Linear probing (and quadratic/double hashing) are open addressing; chaining uses lists.

SECTION B — Practice (full syllabus coverage for revision)

B1. Programming basics

Q31

All 3 exams

C is often called a ____ language:

- (a) Low-level
- (b) Middle-level
- (c) Scripting
- (d) Markup

Answer: (b)

C mixes high-level syntax with low-level (pointer/memory) access, hence middle-level.

Q32

All 3 exams

Which decides the order of evaluation in an expression?

- (a) Indentation
- (b) Operator precedence & associativity
- (c) Line number
- (d) Variable name

Answer: (b)

Precedence and associativity rules determine evaluation order.

Q33

All 3 exams

break inside a loop:

- (a) Skips one iteration
- (b) Exits the loop
- (c) Restarts the loop
- (d) Ends the program

Answer: (b)

break exits the loop entirely; continue skips to the next iteration.

Q34

DSSSB TGT

All 3 exams

In Python, which type is immutable?

- (a) list
- (b) dict
- (c) tuple
- (d) set

Answer: (c)

Tuples (and int/float/str) are immutable; list/dict/set are mutable.

Q35

DSSSB TGT

All 3 exams

Python blocks are defined by:

- (a) Braces {}
- (b) Indentation
- (c) Semicolons
- (d) begin/end

Answer: (b)

Python uses indentation to delimit code blocks.

B2. OOP

Q36

All 3 exams

Exposing only essential features and hiding complexity is:

- (a) Encapsulation
- (b) Abstraction
- (c) Inheritance
- (d) Polymorphism

Answer: (b)

Abstraction hides implementation detail, exposing only what's necessary.

Q37

All 3 exams

Overriding redefines a method in a:

- (a) Same class
- (b) Child class
- (c) Interface only
- (d) Constructor

Answer: (b)

Overriding gives a child class its own version of a parent method (run-time polymorphism).

Q38

All 3 exams

A constructor is used to:

- (a) Destroy an object
- (b) Initialise an object
- (c) Compile code
- (d) Print output

Answer: (b)

A constructor initialises a new object (same name as the class).

Q39

DSSSB TGT

All 3 exams

Which access specifier makes a member accessible only within its class?

- (a) public
- (b) protected
- (c) private
- (d) default

Answer: (c)

private limits access to within the class; protected adds subclasses; public is everywhere.

B3. Linear structures

Q40

All 3 exams

Compared to arrays, linked lists offer:

- (a) $O(1)$ random access
- (b) Dynamic size and cheap insert/delete at a node
- (c) Fixed size
- (d) Contiguous memory

Answer: (b)

Linked lists grow/shrink dynamically and support $O(1)$ insert/delete at a known node.

Q41

All 3 exams

A doubly linked list node has:

- (a) One pointer
- (b) Two pointers (prev & next)
- (c) No pointer
- (d) Three pointers

Answer: (b)

A doubly linked list node stores prev and next pointers.

Q42

All 3 exams

Push and pop operate on a stack at the:

- (a) Front
- (b) Rear
- (c) Top (same end)
- (d) Both ends

Answer: (c)

Stack push/pop happen at the same end (top).

Q43

All 3 exams

A priority queue serves:

- (a) FIFO
- (b) LIFO
- (c) Highest-priority element first
- (d) Random element

Answer: (c)

A priority queue dequeues the highest-priority element regardless of insertion order.

B4. Trees & graphs

Q44

All 3 exams

A leaf node is one with:

- (a) No parent
- (b) No children
- (c) One child
- (d) Two children

Answer: (b)

A leaf has no children.

Q45

All 3 exams

The height of a tree is the:

- (a) Number of nodes
- (b) Longest path from root to a leaf
- (c) Number of leaves
- (d) Degree of root

Answer: (b)

Height = longest root-to-leaf path length.

Q46

GATE-level

Coal India MT

Which self-balancing tree uses rotations to keep height $\sim \log n$?

- (a) BST
- (b) AVL tree
- (c) Linked list
- (d) Heap

Answer: (b)

AVL (and Red-Black) trees rebalance via rotations to stay $\sim \log n$ tall.

Q47

All 3 exams

An adjacency matrix for a graph of V vertices uses space:

- (a) $O(V)$
- (b) $O(V^2)$
- (c) $O(E)$
- (d) $O(1)$

Answer: (b)

An adjacency matrix uses $O(V^2)$ space; an adjacency list uses $O(V+E)$.

Q48

All 3 exams

DFS explores:

- (a) Level by level
- (b) As deep as possible before backtracking
- (c) Only neighbours
- (d) Randomly

Answer: (b)

DFS goes deep along a path, backtracking when needed (stack/recursion).

B5. Hashing

Q49

All 3 exams

The load factor of a hash table is:

- (a) Table size / entries
- (b) Entries / table size
- (c) Number of collisions
- (d) Key length

Answer: (b)

Load factor = entries ÷ table size; higher means more collisions.

Q50

All 3 exams

Chaining resolves collisions by:

- (a) Probing next slot
- (b) Storing a linked list per slot
- (c) Rehashing all
- (d) Ignoring

Answer: (b)

Chaining keeps a linked list of colliding keys at each slot.

Q51

GATE-level

Coal India MT

Double hashing is a form of:

- (a) Chaining
- (b) Open addressing
- (c) Separate lists
- (d) Tree indexing

Answer: (b)

Double hashing is an open-addressing probe sequence using a second hash function.

Coverage & next

✓ What this bank covered

PYQ + Practice across programming fundamentals, OOP, linear structures (array/list/stack/queue), trees/heaps/graphs, and hashing.

GATE-tagged items (recursion/pointer internals, AVL, node-count formulas, double hashing) are Coal-India depth.

Note: original questions modelled on real exam patterns, not verbatim reproductions.